

Syed Muhammad Ammar

🌐Portfolio | ✉syed.m.ammar123@gmail.com | 📞+92 3363058659 | [in LinkedIn](#) | [GitHub](#) | 📍Khi, Pakistan

EDUCATION

• NED University of Engineering and Technology

Bachelor of Engineering in Software Engineering

Nov 2021 – Sep 2025

Karachi, Pakistan

PROFESSIONAL EXPERIENCE

• IOMechs 🌐

Software Engineer

Jan 2026 – Present

- Engineering mobile products, including a [ride-hailing app](#) and an [events-discovery platform](#).
- Owned end-to-end feature delivery across both products. Reduced third-party API calls by 65% through a custom algorithm and improved rendering performance via virtualization and caching.
- Utilized React Native, TypeScript, and Redux Toolkit/RTK Query to build efficient, performant apps.

• MatchMaker GME 🌐

Software Engineer

Apr 2024 – Dec 2025

- Developed several full-stack web and mobile applications for education and academic evaluation.
- Implemented AI systems using LLMs for automated text analysis and content generation.
- Handled authentication, database design, and deployment of back-end and front-end services.
- Optimized database designs to cut data fetching costs by up to 25%.
- Implemented an LLM-as-judge evaluation pipeline to validate 40k+ database items, with human-in-the-loop review across multiple models for quality assurance.
- Developed a chatbot and implemented prompt-engineering-based guardrails to enforce safe outputs.
- Oversaw full product lifecycles while coordinating with stakeholders.

• MetaTalent.ai 🌐

Software Engineering Intern

Jan 2024 – Apr 2024

- Built ~20-screen admin panel with 3 custom dashboards, enabling full platform and user management.
- Developed several skill assessment features, implementing both front-end and back-end functionality.
- Designed RESTful APIs and optimized DB schemas for efficient data communication and management.

• Evincible Solutions 🌐

Software Engineering Intern

Aug 2023 – Sep 2023

- Built an app and several interfaces using power apps and power pages, for smooth property assessments.
- Designed a chatbot for a hotel booking system using Power Virtual Agents, automating reservations.
- Automated business workflows and developed custom components, improving organizational efficiency.

SKILLS

- **Languages:** JavaScript, TypeScript, Python, SQL
- **Frameworks:** Node.js, Express.js, React, React Native
- **Databases:** PostgreSQL, MongoDB, Firebase, Redis
- **AI:** LangChain, RAG, Agentic AI
- **Tools:** Claude Code, Git, Figma, Postman, UML, Latex, Github Actions

ACHIEVEMENTS AND AWARDS

- **Stanford Code in Place:** Selected as Section Leader, taught programming to international students. [\[Link\]](#)
- **GreenCrop.ai:** recognized among the **Best Projects of 2025** by the **UNESCO Chair at NED**. [\[Link\]](#)
- Presented **GreenCrop.ai** at the **Climatech International Conference**, Karachi, Pakistan (2025). [\[Slides\]](#)
- **1st Place, IBA HackFest 2025:** led the winning team in a Shark Tank-style innovation competition. [\[Link\]](#)
- **Regional Runner-up, IEEE YESIST12:** **GreenCrop.ai** ranked 2nd among teams from top universities in a competitive challenge. [\[Link\]](#)
- **Runner-up, CodeSphere'23:** among 100+ participants in a departmental programming contest. [\[Link\]](#)

PROJECTS

• **GreenCrop.ai(AgriSense): AI+IoT-Enabled Precision Farming Platform**

Technologies: *React Native, TensorFlow, LangChain, NestJS, FastAPI, PostgreSQL, LLMs, ESP32, Azure, Hugging face*



- Developed ML models, including a disease detection module (92% acc), recommendation engine (95%).
- Created a decision-support system using generative AI with RAG for localized, context-aware insights.
- Architected an IoT device, integrating sensors, microcontrollers, and solar power for soil monitoring.
- Integrated voice-based AI models to enhance accessibility, expanding usability to millions of farmers.
- Designed 15+ screens, 12+ APIs, and deployed services across Azure and Render for smooth integration.
- Led a 4-person team, managing the product lifecycle and marketing the product by pitching at 5+ events.

• **ResearchRank: AI-Powered Student Ranking System**

Technologies: *ReactJS, JavaScript, LLMs, GitHub, Firebase, Vercel*



- Built a ranking system using publication metrics, personal statement, and recommendations.
- Integrated LLM for automated evaluation, with customizable scoring criteria and weightages.
- Engineered authentication, db schemas, and report generation for streamlined data handling.

• **SoulSupport: A Mental Health Website**

Technologies: *ReactJS, Node.js, Express, Tailwind, LLMs, Stripe, MongoDB, Mongoose ORM, GitHub*



- Built full-stack mental health platform with AI chatbot, video consultations, educational content, online sessions, role-based authentication, and payment integration.
- Architected scalable schemas and optimized RESTful APIs for seamless therapist-patient interactions.
- Integrated frontend designs with backend services and managed production deployments.

• **ScrubHub: Medical Education App**

Technologies: *React Native, Firebase, Next.js, Netlify, Tailwind CSS, LLMs*



- Developed/deployed cross-platform app with personalized learning, and social collaboration features.
- Integrated LLMs to auto-generate 18,000+ questions and architected scalable db schemas.
- Led product strategy and deployment, designing scalable data and UX systems.

• **GoatLang: A Programming Language**

Technologies: *JavaScript, React, Node.js, Express.js, Babel, GitHub, Trello, Vercel*



- Designed a compiler pipeline with lexical analysis, parsing, AST construction, and JS code generation.
- Built a web-based IDE for coding, testing, and learning GoatLang with built-in documentation resource.
- Created an Abstract Syntax Tree visualizer to illustrate how GoatLang programs are processed internally.
- Applied data structures and object-oriented principles to design parsing and language components.

• **Deep Research AI: Multi-Agent Research System**

Technologies: *Flask, Python, LlamaIndex, Groq, Tavily API, JavaScript, HTML, CSS*



- Developed a multi-agent AI system to autonomously conduct topic-based research.
- Implemented the pipeline for question generation, web-based information retrieval, and report synthesis.
- Built a web interface with real-time progress tracking and markdown-formatted search outputs.

PUBLICATIONS

[1] An Analysis of Security Risks and Productivity Impacts of AI-Generated Code [[Link](#)]
AI and Ethics, Springer Nature.

[2] GreenCrop.ai: Smart Solution for Precision Farming, and Resource Efficiency [[Link](#)]
In Proceedings of the 2nd IEEE Humanitarian Technology Conference (KHI-HTC); to appear in IEEE Xplore.

COURSES AND CERTIFICATIONS

- **Coursera(Meta):** HTML and CSS in depth, React Basics & Advanced
- **Jawan Pakistan:** UI/UX Designing & Front-End Web Development
- **Frontend Masters:** Intro to Node.js, API Design in Node.js, Claude Code, Prompt Engineering